Quiz 1

Q determinant of null matrix

ans 0

Q if the order of 3×2 then what should be the order of matrix B, Subtraction is possible.

ans 3×2

Q Two matrix same order can be

ans added and Subtract

Q:- determinant is process of allotting a real number to every

ans square.

Q If the order of matrix A is 3×3 then what should be the order of matrix B such that addition of matrix A and B possible

3×3

Q1

determinant of identity matrix?

ans 1

Q2 If a scalar 2 is multiplied with the matrix then which one following statement is correct?

ans Two 2 will be multiplied with all elements of the matrix

Q3 Given a square matrix A, if there exist a matrix B such that $AB = BA = I$, then A is called the ___ of B

ans multiple inverse

Q2 which of the following matrix have has an inverse.

ans non-Singular matrix

Q1 To add two matrix order must be

MTH 100

10 A square matrix which one diagonal (leading) are zero else where is called a/an ---- matrix.

ans :- Identity matrix

09 A minor of an element in a third order determinant is a ------

ans Second order

08 If B is a matrix whose determinant zero then it inverse. - - -

(a) will be zero (b) cannot determined (c) will be negative (d) will be positive.

07 if matrix have A an order 3×5 matrix B order 5×3, order of AxB is

ans => 3×3

06 A and B are any two matrices of order m and respectively and m > n and p < q for the product condition m n p q AB hold.

(a) n=p (b) n>p (c) n<p

Quiz 07

1) Identify the sequence Arithmetics of 8, 13, 18, 23

(a) Arithmetic (b) Geometric

Ans (c)

2) Explicit formula for the sequence with term 2, 7, 12, 17 is

(a) $5n+3$ (b) $5n-3$ (c) $n+2$

Ans (d) $n-3$

3) For a geometric sequence if 2nd term is equal to 4th term then value of common ratio is.

(a) -1 only (b) 1 only (c) 0

Ans (d) $+1$ or -1

4) If the sum of first n natural numbers is $n(n+1)/2$ the first- average will be.

(a) $n/2$ (b) $(n-1)/2$ (c) $(n+1)/2$

Ans (d) $n(n+1)/2$

5) If the first term $a=1$ common ratio $r=1/2$ then which term of geometric sequence is $\frac{1}{8}$

(a) 5 (b) 6 (c) 3

Ans (d) 4

Q3)

5. Sum of how many term of an arithmetics series will be zero provided that 1st term = 20, common difference = 20?

(a) 20 (b) 10 (c) 3

Ans (d) 2

7. The 9999th term of the sequence (-1, 1, -1, 1, -1, 1 ....) is

(a) -1 (b) 9999

Ans (c) +-9999 (d) Undefined

9. For a geometric progression if first term = 7 and common ratio = 3 then the 4th term is ...

(a) 121 (b) 144 (c) 189

Ans (d) 16

10. For an arithmetic series, if $a_1 = -3$, $d = 2$ and $S_n = 21$ then the number of term "n" in the series are

Ans (a) 7 (b) -7 (c) 10

Quiz 02

10. The explicit formula when $a_1 = 6$ and $d = 2$ is ...

(a) $a_n = 6 + 2(n+1)$ (b) $a_n = 6 + 2n$

Ans (c) $a_n = 6 - 2(n-1)$ $a_n = 6 + 2(n-1)$

11. For the sequence 5, 10, 20, 40 ... eight term will be ...

(a) 640 (b) 19 (c) 70

Ans (d) 90

12. If first term of geometric sequence is 243 and sixth term is 1, then the common ratio will be.

(a) 3 (b) 1 (c) 1/3

Ans (d) -1

13. For a sequence whose nth term is "$(-200/n)$" then its 200th term is

Ans (a) 200 (b) 1

(c) -1 (d) -4000

Quiz 07

14) If $C_n = n(n+1)$ for all integers $n \geq 1$ then which of the following is the first three term of a sequence..

(a) 0, 2, 6 .. (b) 2, 6, 12 ..

Ans (c) 1, 2, 3 .. (d) 1, 2, 6

15) Which of the following is an arithmetic sequence.

Ans (a) 1, 4, 7 .. (b) 1, 4, 9, ...

(c) 1/2, 1/4, 1/6 (d) 2, 4, 8, ...

16) Sum of first 998 term of sequence $(-1, 1, -1, 1 \ldots)$ is?

(a) 0 , (b) 1 (c) -1 , (d) Undefine

17) Sum of first 2000 term of the sequence $(-1, 1, -1, 1 \ldots)$ is

Ans (a) -1 (b) 1 (c) 0 (d) Undefine

02

18) If the common ratio = 3, for the following geometric sequence $(x-1), (x-1)(x-1), (x-1)(x-1)(x-1)$ --- then the value of x is.

Ans (a) 3 (b) 4 (c) 2 (d) 1

19) For a geometric sequence if nth term = 32, first term = common ratio = 2 then the value of 'n' is —

Ans (a) 2 also! (b) 3 (c) 5

(d) 1

20) In first term = a = 2 common difference = d = 4 and term number = n = 10 then what is value of 10th term in arithmetic sequence?

Ans (a) 40 (b) 39 (c) 38

(d) 37

Quiz 02

21. For the sequence of natural number, if sum of 'n' natural number = summation $(n) = 21$ then value of 'n' is ---

Ans (a) -7 (b) -6 (c) 6 (d) 7

Quiz 02

23) The first term of sequence whose nth term = $2n > 2$ is - - -

Ans (a) 2 (b) 4 (c) 6 (d) 1

23) Sum of the first 20 term of the following series:
$1+2+3+4 - - -$ is

Ans (a) 55 (b) 200 (c) 220
(d) 210

24) Sum of the geometric series: $1 + (1/2) + (1/4) + (1/8) + \cdots$

Ans (a) 2 (b) 3/2 (c) 1
(d) 5/2

25) [illegible]

Quiz 02

25) if the difference b/w successive term of a series is a constant, then the series is said to be a/an --- series?

Ans (a) Arithmetic (b) Geometric
(c) Harmonic (d) none

26) For a geometric sequence $(1, b, b, b, bbb\ldots)$ if its common ratio is 3 then value of b is

Ans (a) -3 (b) 3
(c) 1 or -1 (d) 2

27) Identify the sequence as Arithmetic, Geometric or neither
8, 10, 20, 40

Ans (a) Arithmetic (b) Geometric

MTH 100 Quiz 03

1) $\sin^2\theta + \cos^2\theta$ = ...

(a) -1 (b) 0 (c) $1/2$ (d) 1 ✓

2) $\cos(-\theta)$ = ...

(a) $\sec(\theta)$ ~~(b) $\cot(\theta)$~~ (b) $\cos\theta$

(c) $-\cos\theta$ (d) $-\cos(-\theta)$

3) $\sin(-\theta)$ = ...

(a) $\sin\theta$ (b) $-\sin\theta$ (c) $\cos\theta$

(d) $\cos(-\theta)$

4) The sum formula for $\tan(A+B)$ is .....

$$\frac{\tan A + \tan B}{1 - \tan A \tan B}$$

5) All the trigonometric function have positive value in ..... quadrant

(a) 1st (b) 2nd

Quiz 3

6 ⇒ The function $Y = \sin 4x$ has period

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{4}$ (d) 2π

7 ⇒ The domain of inverse tangent function is - - - -

(-infinity; infinity)

8 ⇒ If I adde) -5 to each value coordinate point (-23), what would be new coordinate?

(a) (7,2) (b) (-7,-2) (c) (3,2)

(d) (10, -15)

9 ⇒ The smallest interval function repeat it self is called

(a) Slope (b) Period (c) Range

(d) Graph

10 ⇒ The point where the x- and y- axis cross (0,0)

(a) intercept (b) Slop

Q3

11. Using the distance formula find the distance b/w the two points A(1,2), B(4,6)

(a) -5 (b) 5

(c) 13 (d) 7

12. Smallest interval ~~function~~ for which the trigonometric function repeat it self is called

(a) amplitude (b) Period

(c) angle (d) range

13. Power reducing formula $\cos^2 A$ is ---

(a) $\frac{2 + \tan A}{1 + \cos^2 A}$ (b) $1 - \cos^2 A$

(c) $\frac{\cos 2A + 1}{2}$ (d) $\frac{1 - \cos 2A}{2}$

14. Sin θ is positive in the first and the --- quadrant

Second

(Homework pg 37)

15 ⇒ The ---- form of a trigonometric function is the height from the mean (or the rest) value of the trigonometric function to its maximum or minimum value

Amplitude (Hawoni pg 40)

16 ⇒ The difference formula for $\sin(A-B)$ is

(Hawoni pg 48) $\sin(A-B) = \sin A\cos B - \cos A\sin B$

17 ⇒ Equation that is not ~~identity~~ identity is called a(an) ---- equation

(a) Identical (b) linear
(c) Conditional (d) transcendental

18 ⇒ $\cos\theta$ is positive in the --- and the third quadrant

First (Hawoni pg 38)

Q no 3

19) The range of cosine function is b/w -1 and 1

(a) True (b) False

20) Cos θ is negative in the second and ---- quadrant

Fourth (Ans pg 38)

21) The amplitude of the function is the height from the rest value to the extreme value

(a) True (b) False

22) Equation of circle centered at (0, -5) and radius 5 is sq.m-85 is

(a) $S(x^2) \cdot 2(y+5)^2 = 8$ (b) $S\{x^2 + (y+5)^2\} - 64$

(c) $S\{x^2 + (y+5)^2\} - 645$

(d) $S(x^2) \cdot (y+5)^2 = 8$

Ques 23

23) The center of circle having equation $(x-2)^2 - (y-5)^2 - 36 = 0$ is at-

(a) $(-2, -5)$ (b) $(2, -5)$

(c) $(-2, 5)$ (d) $(2, 5)$

24) Using the distance formula find the distance b/w the two point- A $(4, 7)$, B $(3, -2)$

(a) 7 (b) 19

(c) $\sqrt{82}$ (d) 20

25) $\cos(180 - \theta) = \ldots$

$-\cos\theta$ (Identity Pg 82)

26) $1 + \cot^2\theta = \ldots$

$\csc^2\theta$ (Identity Pg 97)

27) The range of $y = \arccos x$ is

$[-1, 1]$ $[0, \pi]$

Quiz 03

28) The mid point of the line segment with the end point $(-4, 4), (-2, 2)$ is

(a) $(-6, 6)$ (b) $(2, 2)$

(c) $(-3, 3)$ (d) $(-2, -2)$

29) Equation of a circle centered at $(0, -5)$ and radius $\sqrt{8}$ is

$$x^2 + (y+5)^2 = 8$$

30) Tangent function is periodic but it's periodic is ___ degree

(a) 60 (b) 90

(c) 180 (d) 360

31) The domain of $y = \arcsin x$ is

(a) $[-1, 1]$ (b) $(-\pi, \pi)$

(c) $[-\infty, \infty]$ (d) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Quiz 03

2) $\cos 2A =$

(a) $2\cos^2 A - 1$ (b) $1 - \cos^2 A$

(c) $\cos^2 A + \sin^2 A$ (d) $2 \sin A \cos A$

3) The sin function is periodic with periodic 360 degree and range b/w - --

(a) 0 and -1 (b) 0 and 1

(c) -1 and 0.5 (d) -1 and 1

4) $\sin\left(-\frac{\pi}{4}\right) =$ - --

(a) $-\sin\left(\frac{\pi}{4}\right)$ (b) $\sin\left(\frac{\pi}{4}\right)$

(c) $\text{cosec}\left(\frac{\pi}{4}\right)$ (d) $-\sec\left(\frac{\pi}{4}\right)$

MTH 100

18) For individual observation, reciprocal of arithmetic mean is called Harmonic mean

(04) MTH 100

14) Budget of a household can be represented effectively through which of following chart.
(a) Bar (b) Frequency
(c) Pie (d) Histogram

15) if the mean, median mode of two data sets are equal then
(a) both data set should be equal
(b) both data set not necessarily equal
(c) both data set should not be equal
(d) none of them

16) when data is arranged, middle value in set of observation is identified as
(a) median (b) mean
(c) variance (d) standard deviation

17) median of the following data
4, 5, 6, 7, 8, 9, 10
(a) 6 (b) 6.5
(c) 7 (d) 7.5

(3)

MTH 100

10) If the cumulative frequencies are plotted against upper class boundaries of the corresponding class
(a) True (b) False

11) Arithmetic means 10 and number of observation are 20 then sum of all value is.
(a) ~~20~~ 20 (b) 100
(c) 200 (d) None of them

12) A quantative variable which can take any in a given range is called
(a) Simple (b) discrete
(c) Contonous (d) none of them

13) If a variable cannot take numerical value then variable is called
(a) Constant (b) discrete
(c) Contonous (d) Quantative

2) MTH 100

4) Arithmetic mean 2, 4 and 6
(a) 3 (b) 4
(c) 5 (d) 6

5) The central tendency is measured by
(a) range (b) average
(c) dispersion (d) deviation

7) pieces of numerical and non-numerical information are called data.
(a) True (b) False

8) The mode of the list (1, 2, 2, 3, 3, 3, 4) is
(a) 2 (b) 2.5
(c) 3 (d) 4

9) Relative frequency calculated by dividing the frequency of each interval. (a) length of interval
(b) sum of all interval
(c) total number of observation
(d) None of them

(1)

MTH 100 Quiz 04

1) Frequency Density =
(a) Cumulative Frequency / Relative
(b) Cumulative / Class width
(c) Frequency / Class width
(d) Class width / Frequency

2) Which of the following is not used to measure the central tendency.
(a) Arithmetic (b) Mode
(c) Median (d) Standard Deviation

3) A bar chart which represent continuous data is also
(a) Histogram (b) Pie chart
(c) Discrete chart (d) None of them

4) Method used to compute average or central value of collected data is consider.
(a) measure of central tendency
(b) measure of negative skewness
(c) negative variation
(d) Positive variation.